

```
mutex = 1
full = 0
empty = 10
x = 0
def producer(): 1 usage
    global mutex, full, empty, x
    mutex -= 1
    full += 1
    empty -= 1
    x += 1
    print(f"\nProducer produces item {x}")
    mutex += 1
def consumer(): 1 usage
    global mutex, full, empty, x
    mutex -= 1
    full -= 1
    empty += 1
    print(f"\nConsumer consumes item {x}")
    x -= 1
    mutex += 1
print("1. Press 1 for Producer")
print("2. Press 2 for Consumer")
print("3. Press 3 for Exit")
```

```
while True:
    n = int(input("\nEnter your choice: "))
    if n == 1:
        if mutex == 1 and empty != 0:
            producer()
        else:
            print("Buffer is full!")
    elif n == 2:
        if mutex == 1 and full != 0:
            consumer()
        else:
            print("Buffer is empty!")
    elif n == 3:
        print("Exiting...")
        break
    else:
        print("Invalid Choice")
```

1. Press 1 for Producer
2. Press 2 for Consumer
3. Press 3 for Exit

Enter your choice: 2

Buffer is empty!

Enter your choice: 3

Exiting...

Process finished with exit code 0